

Diagnostic report -- blind - normal (0.000)

Overall status: CRITICAL Generated at 2026-08-30T18:05:39.492985+00:00 | Cycle (most recent signal): 2026-08-30T18:05:39.485989+00:00

Executive summary

Data drift: 2/8 feature(s) drifted (severity critical). Target drift (confirmed): PSI=0.000, severity none. Performance: severity critical, cause = model_weights. Alerts: 3 critical, 1 warning. 3 remediation action(s) pending human approval.

Probable root cause identified (cross-agent corroboration)

Probable root cause identified for the performance degradation

Performance degradation (model_weights) coincides with confirmed drift on Glucose, feature(s) with direct SHAP impact: likely causal link.

Recommended action: Prioritize fixing (data or pipeline) Glucose before any blind retrain -- retraining without fixing the cause would reproduce the same problem next cycle.

1. Data / Feature Drift

Overall severity: **critical** -- 2/8 feature(s) drifted (25.0%).

Feature	Type	PSI	Wasserstein (norm)	Severity	Drifted
Pregnancies	numeric	0.0005	0.0052	none	no
Glucose	numeric	0.1022	0.2253	critical	yes
BloodPressure	numeric	0.0003	0.0095	none	no
SkinThickness	numeric	0.0003	0.0080	none	no
Insulin	numeric	0.0004	0.0276	none	no
BMI	numeric	0.1182	0.2406	critical	yes
DiabetesPedigreeFunction	numeric	0.0004	0.0097	none	no
Age	numeric	0.0005	0.0025	none	no

2. Target Drift

Status: **confirmed** | n_samples = 740

PSI = 0.0000 | severity = **none** | drifted = no Wasserstein distance (normalized) = 0.0267 Reference mean = 0.3490 -> current mean = 0.3757 (reference std = 0.4766 -> current std = 0.4843)

3. Performance Drift

Status: **labeled** | severity: **critical** | remediation_type: model_weights | window = 500 samples y_proba coverage over the window: 100.0%

Performance metrics

Metric	Baseline	Current	Delta	Std score	Degraded	Contributes to severity
accuracy	0.9349	0.9060	-0.0289	-3.09	yes	yes
precision	0.9290	0.7180	-0.2110	-21.21	yes	yes
recall	0.9275	0.9129	-0.0146	-1.51	no	yes
f1	0.9282	0.7725	-0.1557	-16.77	yes	yes
recall_class_1	0.9030	0.9211	+0.0181	+1.41	no	yes
auc_roc	0.9829	0.9469	-0.0361	-3.67	yes	yes
auc_pr	0.9701	0.6857	-0.2844	-29.32	yes	yes

Calibration metrics

Metric	Baseline	Current	Delta	Std score	Degraded	Contributes to severity
log_loss	0.1954	0.2695	+0.0741	+7.08	yes	yes
brier_score	0.1041	0.1527	+0.0487	+7.46	yes	yes
ece	0.0591	0.0405	-0.0187	-3.40	no	yes

Alerts raised this cycle (4)

[CRITICAL] High model-impact drift on 1 feature(s) (source: data_drift)

Affected features: Glucose. - Glucose: 'Glucose' is at the 88th percentile of SHAP importance among tracked features (threshold=75th, raw importance=0.5135): the drift directly affects predictions.

Recommended action (expert): FINAL VERDICT -- retrain: batch is clean (no missing-value spike, no new category) on the drifted feature(s), so this is a genuine population shift, not a data-quality artifact. Prioritize Glucose (direct model impact per SHAP). Check for a root_cause alert this cycle before scheduling the retrain.

Remediation type: *retrain*

[CRITICAL] Real model performance degradation (source: performance_drift)

Degraded metrics this window: accuracy, precision, f1, auc_roc, auc_pr.

Recommended action (expert): Retrain recommended. Check first whether a root_cause alert was raised this cycle: if so, fix that feature/pipeline before retraining, or the same issue will recur.

Remediation type: *retrain*

[CRITICAL] Probable root cause identified for the performance degradation (source: root_cause)

Performance degradation (model_weights) coincides with confirmed drift on Glucose, feature(s) with direct SHAP impact: likely causal link.

Recommended action (expert): Prioritize fixing (data or pipeline) Glucose before any blind retrain -- retraining without fixing the cause would reproduce the same problem next cycle.

Remediation type: *retrain*

[WARNING] Drift to monitor on 1 feature(s) (source: data_drift)

Affected features: BMI. - BMI: 'BMI' has low SHAP importance (percentile 50) and isn't correlated with an influential feature: likely negligible model impact despite the statistical drift.

Recommended action (expert): No immediate action: risk is indirect (correlated with a key feature) or undetermined (missing SHAP/correlation data). Re-evaluate next cycle.

Remediation type: *monitor*

Remediation actions (4)

Type	Urgency (SLA)	Status	Approval required	Automatable	Source alert
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retrain	urgent (within 24h) -- superseded by 'Probable root cause identified for the performance degradation'	skipped	yes	yes	High model-impact drift on 1 feature(s)
monitor	normal (next working cycle)	executed	no	yes	Drift to monitor on 1 feature(s)
retrain	urgent (within 24h) -- superseded by 'Probable root cause identified for the performance degradation'	skipped	yes	yes	Real model performance degradation
retrain	urgent (within 24h)	pending	yes	yes	Probable root cause identified for the performance degradation

Generated automatically by the drift monitoring pipeline.